



SYNC SCREEN TO SIGHT



PROBLEM STATEMENT:

Millions of people struggle to read digital content comfortably due to varying visual impairments such as myopia, hyperopia, presbyopia, and eye strain.

- Glasses are personalized but static.
- Digital devices do not automatically adapt to different users.
- Shared devices (schools, offices, libraries) cannot adjust for individual vision needs.
- Poor readability reduces learning efficiency and productivity.

There is no intelligent system that dynamically adapts digital screens based on individual visual capability in real time.



OUR SOLUTION

Sync Screen to Sight is an AI-powered adaptive reading assistant that:

- Assesses visual acuity using a Snellen-based digital test
- Generates a personalized vision profile
- Tracks gaze, pupil movement, and reading patterns
- Dynamically adjusts text size, spacing, font, contrast, and brightness
- Supports multiple users on the same device without manual reset
- Continuously improves using AI-based learning

It transforms any digital screen into a personalized smart display.

TARGET USERS

- Students in schools & colleges
- Office professionals
- Elderly users
- People adapting to new prescriptions
- Public device users (labs, libraries, kiosks)

IMPACT

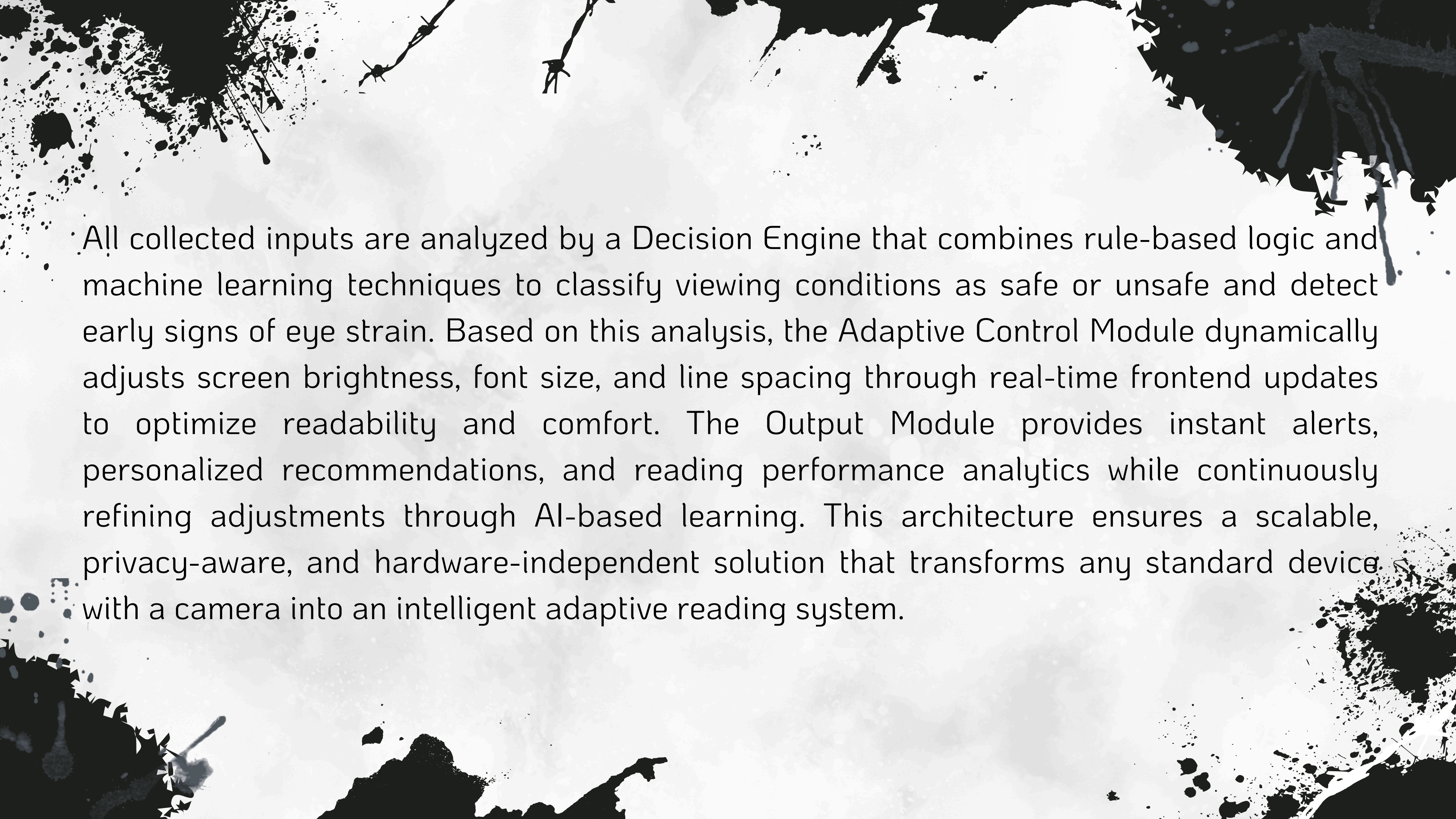
- Reduces eye strain & fatigue
- Improves reading speed & comprehension
- Promotes digital accessibility & inclusivity
- Minimizes dependency on corrective lenses in digital environments
- Enables seamless multi-user experience

Potential use in:

- Education | Workplaces | E-readers | Public digital systems

TECHNICAL APPROACH

“Sync Screen to Sight – AI-Based Adaptive Reading Assistant” is built on a modular AI-driven architecture that integrates a web-based frontend with a Django backend and real-time computer vision processing. The frontend, developed using HTML, CSS, and JavaScript, captures user interactions and live video streams through browser APIs and communicates with the backend via restful services. The system begins with a Vision Test Module that conducts a digital Snellen chart assessment to estimate visual acuity and approximate diopter values, generating a personalized vision profile stored securely in the database. In parallel, the Camera Processing Module uses OpenCV and MediaPipe to perform real-time face detection, eye landmark tracking, gaze analysis, and blink rate monitoring. A Distance Calculation Module then converts pixel-based facial measurements into real-world distance (in centimeters) using geometric estimation formulas to continuously monitor the user’s position relative to the screen.



All collected inputs are analyzed by a Decision Engine that combines rule-based logic and machine learning techniques to classify viewing conditions as safe or unsafe and detect early signs of eye strain. Based on this analysis, the Adaptive Control Module dynamically adjusts screen brightness, font size, and line spacing through real-time frontend updates to optimize readability and comfort. The Output Module provides instant alerts, personalized recommendations, and reading performance analytics while continuously refining adjustments through AI-based learning. This architecture ensures a scalable, privacy-aware, and hardware-independent solution that transforms any standard device with a camera into an intelligent adaptive reading system.

INNOVATION

- Unlike glasses, this is software-based and adaptive
- Works for unlimited users on the same device
- Combines AI + Computer Vision + Accessibility
- Provides real-time dynamic personalization

This is not just a reading app — it is an AI-powered visual accessibility system.

FEASIBILITY

- Uses existing device cameras
- No extra hardware required
- Scalable as a browser extension, mobile app, or OS-level feature
- Easily integrable with e-learning platforms